§6. Galois Theory of Simple Systems

The sharpened commutation theorem of §5, 4 expresses the Galois theory of simple systems with respect to the center as ground domain. We consider first division rings, and then simple systems in general.

1. Galois theory of division rings of finite rank over the center. The Galois group $\mathfrak{G}$ of A is defined as the group of all automorphisms extending the identity of the center P, hence, by §5, 2, as the group of all inner automorphisms. Thus

$$\mathfrak{G} \simeq A^*/P^*,$$

where A^* and P^* denote the multiplicative groups of nonzero elements of A and P. Subgroups $\mathfrak{H}$ of $\mathfrak{G}$ therefore correspond one-to-one to subgroups H^* of A^* containing P^*, by

$$\mathfrak{H} \sim H^*/P^*.$$

A subgroup $\mathfrak{H}$ of $\mathfrak{G}$ is called closed if, after adjoining zero, H^* becomes a division ring H. The fact that C admits the group $\mathfrak{H}$ is equivalent to saying that C commutes elementwise with H. Hence the commutation theorem of §5, 4 becomes the following.

Main theorem of the Galois theory of noncommutative division rings. The division rings C between P and A and the closed subgroups $\mathfrak{H}$ of $\mathfrak{G}$ correspond bijectively in such a way that C is the full invariant division ring of $\mathfrak{H}$, and $\mathfrak{H}$ is the full invariant group of C.

2. Galois theory of simple systems. If A_f is a simple system with center P, the Galois group $\mathfrak{G}$ is again the group of inner automorphisms, hence isomorphic to

$$A_f^*/P^*,$$

where now A_f^* denotes the multiplicative group of the regular elements of A_f. A subgroup $\mathfrak{H} \sim H^*/P^*$ of $\mathfrak{G}$ is called simple-closed if the subring H generated by H^* is a simple system and if H^* is the set of all regular elements of H. The passage from the commutation theorem to Galois theory is supplied by the following lemma.

Lemma. Every simple system in A_f containing P is generated by the group H^* of its regular elements. This is clear if P has infinitely many elements: the "general element" formed with indeterminates is regular, and by suitable specialization one obtains regular basis elements over P. The lemma holds in general by Shoda.[1]

By the lemma, commutation of S with a simple system $\mathfrak{S}$ is again equivalent to saying that S admits the automorphisms induced by the regular elements of $\mathfrak{S}$. Thus the commutation theorem of §5, 4 becomes also here the following.

Main theorem of the Galois theory of simple systems. The simple systems S in A_f containing P and the simple-closed subgroups $\mathfrak{H}$ of $\mathfrak{G}$ correspond bijectively in such a way that S is the full invariant domain of $\mathfrak{H}$, and $\mathfrak{H}$ is the full invariant group of S.

3. Proof by means of the extension principle. For the case of a division ring I give a second proof of the main theorem. It rests on the principle of extending isomorphisms, i.e. representations of first degree, and since it uses no count of ranks it makes fewer finiteness assumptions. In particular it shows in what direction a transfer to division rings S of infinite rank will be possible. The proof also does not use the fact that there is only one irreducible reciprocal representation class; it therefore applies exactly in the same way in the commutative case, §8, 2. I first state some lemmas.

[1] Shoda, work cited in note 3); there the introduction of indeterminates is avoided.

Hypotheses. The simple system S over P is to admit a reciprocal representation of first degree in A, and is therefore a division ring. Here A may have finite or infinite rank over its center P. A decomposition of S_A into n simple operator-isomorphic right ideals, as in §4, 3, is given by

$$S_A = r_1 + \cdots + r_n = e_1 S_A + \cdots + e_n S_A = e_1 A + \cdots + e_n A;$$

the reciprocal representation generated by e_i is therefore defined by

$$e_i \sigma = e_i \sigma_i.$$

Lemma 1. The n reciprocal representations generated by $e_1, \ldots, e_n$ are distinct, hence are distinct representations of the same representation class. Thus S has at least as many distinct representations as its rank.

To prove this, we extend the representations from S, as at the end of §2, to correspondences of S_A which are operator-homomorphic over A. Here these correspondences are defined by

$$e_i w = e_i \omega$$

with $\omega \in A$, for each $w \in S_A$. If e_i and e_j, $i \neq j$, generated the same representation, one would have $e_i w = e_j w$ for every $w \in S_A$. This is impossible because $e_i e_i = e_i$ and $e_j e_i = 0$.

Lemma 2. Let T be a division ring between P and S, and let s be the rank of S over T. Then every reciprocal isomorphism of T into A admits at least s different extensions.

Let the reciprocal isomorphism, i.e. the reciprocal representation of T in A, be mediated by a decomposition

$$T_A = E_1 T_A + \cdots + E_h T_A = E_1 A + \cdots + E_h A, \qquad E_i t = E_i \tau,$$

where the E_i are idempotents. This is no restriction, since a representation generated by a basis element $E_i \alpha$ is also generated by $\alpha^{-1} E_i \alpha$, hence by an idempotent. Right multiplication by S gives

$$S_A = E_1 S_A + \cdots + E_h S_A.$$

The $E_i S_A$ all have rank s over A. Indeed the rank of $E_i S_A$ is at most s, as follows by substituting a T-basis into S, using the commutativity of S and A:

$$S = T w_1 + \cdots + T w_s,$$

$$E_i S_A = E_i T_A w_1 + \cdots + E_i T_A w_s = E_i w_1 A + \cdots + E_i w_s A.$$

Since the sum S_A has rank $n = hs$, every $E_i S_A$ has exactly rank s. Thus

$$E_i S_A = r_{i1} + \cdots + r_{is} = e_{i1} A + \cdots + e_{is} A,$$

where the s representations generated by $e_{i1}, \ldots, e_{is}$ are all distinct by Lemma 1. They are all extensions of the representation of T generated by E_i, since $E_i t = E_i \tau$ implies

$$(e_{i1} + \cdots + e_{is}) t = (e_{i1} + \cdots + e_{is}) \tau$$

and hence $e_{ij} t = e_{ij} \tau$, by the direct decomposition.

Definition. The partition $\mathfrak{I}_T$ of the reciprocal isomorphisms $\mathfrak{I}$ from S into A, induced by T, is defined as follows: precisely those isomorphisms in $\mathfrak{I}$ are regarded as equivalent which induce the same isomorphism on T.

Main theorem. If $\mathfrak{I}_T$ is the partition induced by T, then T is a maximal subfield relative to this partition.

For if Z is an intermediate division ring between T and S, of degree l over T, then by Lemma 2 every class of $\mathfrak{I}_T$ splits into at least l classes of $\mathfrak{I}_Z$. The transition from this form of the theorem to the usual one arises by multiplying the elements of the set $\mathfrak{I}$ by the inverse of any fixed one of them; the set then becomes the automorphism group, and a class in $\mathfrak{I}_T$ becomes the invariant group of T. The statement that closed subgroups are full invariant groups is included in this: for $\mathfrak{H} \sim H^*/P^*$, one simply puts the division ring H in place of the intermediate division ring T.

§7. The Class of Similar Algebras. Splitting Fields

From now on only division rings $A, \overline{A}$ and matrix rings A_f of finite rank over their center P will occur. Thus they are simple normal algebras over P, and will be called algebras over P, or simply algebras; $A, \overline{A}$ are then division algebras. All A_f with the same associated A are collected into one class of similar algebras A, A_f. Isomorphic A's are identified.

1. The group of algebra classes. If A, B are algebras over P, then the same is true of their product, the direct product over P; for by the end of §3,

$$A_f \times_P B_g = C_h,$$

where C is a uniquely determined, up to isomorphism, division algebra with center P. Multiplication of A_f, B_g by matrix rings over P shows that this multiplication is uniquely determined for the classes, which therefore form a multiplicatively closed, commutative system. Moreover R. Brauer's theorem holds: the classes of similar algebras form an abelian group under direct product. The system has an identity class, consisting of matrix rings over P, that is, of the algebras similar to P. Also each class (A) has the inverse $(\overline{A}) = (A)^{-1}$, consisting of the algebras similar to $\overline{A}$, where $\overline{A}$ is reciprocally isomorphic to A. This follows from the commutation theorem §5, 3 by the specialization $S = A$. For A can be embedded irreducibly in itself; one obtains $B = P$ and $A_A = P_f$.[2]

2. Splitting fields of an algebra class. The theory of splitting fields again rests entirely on the principle stated at the beginning of §5: the commutative extension fields are represented in A or in $\overline{A}$. We first need the following lemma.

Lemma. If A is a division ring with center P, and Z is a finite commutative extension field of P, then A_Z is simple with center Z. For by §4, 1 this holds for the system $Z_{\overline{A}}$, reciprocally isomorphic to A_Z. Thus for every class (A) of algebras over P similar to A, the field Z gives an extension class $(A)_Z$ of simple algebras over Z similar to A_Z.

The associated division algebra D of an extension class follows immediately from the commutation theorem of §5, 3.

Theorem on the associated division algebra of an extension class. Let $(A)_Z$ be an extension class, and let Z denote an irreducible embedding, i.e. representation, of Z in A_f. Then

$$(A)_Z = (D),$$

where D is the set of all elements of A_f which commute elementwise with Z. One only has to replace the simple system S in §5, 3 by Z, observing that Z_A and A_Z are reciprocally isomorphic. If Z is embedded reducibly, the totality of elements commuting elementwise with Z is a matrix ring over D.

A commutative extension field Z of P is called a splitting field of the class (A) if the extension class $(A)_Z$ splits completely, that is, is equal to the identity class over Z, the class of all matrix rings over Z.

Theorem on the characterization of splitting fields. A finite commutative extension field Z of P is a splitting field of the class (A) if and only if its irreducible embedding in A_f gives a maximal commutative subfield of A_f.

Indeed, the property that Z be a splitting field of (A) is equivalent to $(D) = (Z)$. Hence, by the theorem on the associated division algebra, the irreducible embedding Z must be a maximal commutative subfield. If this condition is fulfilled, then necessarily $D = Z$, since adjoining any $a \in D$ to Z gives a commutative extension field.

Remarks. 1. It follows at the same time that a maximal commutative subfield Z, when irreducibly embedded, generates a maximal commutative subring. For the elements commuting elementwise with Z form a division ring.

2. The theorem also gives the existence of splitting fields, since the associated division algebra A certainly has maximal commutative subfields.

[2]The finer theorems on the group of algebra classes use the theory of factor systems; that will not be taken up here. Notice that up to this point no considerations concerning commutative fields have occurred. For example, the fact that the rank $(A : P)$ is a square has neither been used nor proved.

3. Rank relations, index, infinite commutative extensions. The rank statement from §5, 4 first gives the known fact that the rank of a simple algebra over its center is a square. For if Z is maximal commutative in A, then, since every embedding in A is irreducible,

$$(Z:P)(Z:P) = (A:P),$$

hence

$$(A:P) = m^2, \qquad (Z:P) = m,$$

where m is Schur's index; it is also the degree of all splitting fields embeddable in the division algebra A itself. For the degree n of a general splitting field one obtains

$$n = mr,$$

for instance from $(A_r : P) = m^2r^2$, or also from the rank relation §4, 3, since the index m is also the number of simple components of A_Z, which becomes a matrix ring over Z of rank m^2. More generally, if $A_Z \simeq D_l$, if L is the irreducible embedding of A in A_r, and if d^2 is the rank $(D:L)$ of the division algebra D over its center Z, then

$$l\,d = mr.$$

Indeed, by §5, 4 and the theorem on the associated division algebra from 2.,

$$(A_r : P) = (L:P)(D:P),$$

so

$$m^2r^2 = l^2d^2.$$

From the existence of splitting fields follow the facts for infinite commutative, algebraic or transcendental, extension fields Ω of P: the extension class $(A)_\Omega$ is a class of simple algebras with center Ω. If in particular Ω is algebraically closed, then A_Ω is a matrix ring of degree m. Thus the index is equal to the "absolute number of components." For if Z is a splitting field of A and Ω' is the compositum of Ω and Z, then $A_{\Omega'}$ is a matrix ring over Ω', hence has no radical and has center Ω'. Therefore A_Ω also has no radical and its center is Ω; any element in the center of A_Ω not belonging to Ω would also lie in the center of $A_{\Omega'}$. Hence A_Ω is a simple algebra over Ω, and certainly a matrix ring over Ω if Ω is algebraically closed.[3]

4. Existence of separable splitting fields. Every class (A) has separable splitting fields, indeed such that can be embedded in A itself.[4]

Let the ground field P have characteristic p, and let the index be

$$m = sp^f$$

with s prime to p. If Z is a splitting field of A of degree m, and Z_1 is the extension of first kind contained in it, so that

$$(Z : Z_1) = p^f,$$

then the index of $A_{Z_1} \sim D$ is p^f. We prove the existence of a separable extension field Z_2 of Z_1 such that the index of D_{Z_2} is a proper divisor of p^f. Now D_Ω, with Ω algebraically closed, has no radical by 3.; the reduced discriminant of D therefore does not vanish. Hence there is at least one element $d \in D$ with nonzero reduced trace. This d cannot already lie in the ground field Z_1, because the trace of every element $\alpha \in Z_1$ is $p^f\alpha$, hence vanishes. Thus $Z_2 = Z_1(d)$ is a proper, indeed separable, extension field; the index of D_{Z_2} is a proper divisor of p^f. Repeating this finitely many times gives a separable splitting field of (A) whose degree m agrees with the index of A.

§8. Splitting Fields, Decomposition Fields, and Galois Theory in the Commutative Case

In order to pass from the splitting fields of systems simple over their center to those of arbitrary simple systems, one still has to develop the splitting theory of the center; a Galois theory parallel to §6, 3 is to be added to it. All fields and systems occurring in this paragraph are commutative.

[3] In general there are also "minimal" splitting fields, i.e. such that no proper subfield is a splitting field, of all possible degrees. Compare Brauer–Noether, the work cited in note 2).

[4] Compare G. Köthe, Über Schiefkörper mit Unterkörpern zweiter Art über dem Zentrum, Journ. f. Math. 166 (1932), pp. 182–184. The present, much simpler proof goes back to a remark of M. Zorn.

1. Splitting and decomposition fields of commutative simple systems. Let the commutative system Z be simple over P, hence a field, and let Ω be an algebraically closed extension field of P. It is known that Z_Ω remains a system without radical if and only if Z is separable, i.e. an extension of first kind, over P. For only then does it have as many isomorphic maps of first degree into Ω as its rank over P, hence as many distinct representations, which in this case coincide with representation classes.

The possibility that a radical occurs in Z_Ω, so that a direct decomposition into absolutely simple components need not take place, leads to the following definitions. An extension field Λ of P is called a splitting field if Z_Λ splits into composition factors of first degree; in other words, if all absolutely irreducible representations already lie in Λ. An extension field T of P is called a decomposition field if Z_T splits off at least one composition factor of first degree; in other words, if at least one absolutely irreducible representation of Z exists in T.

Splitting and decomposition fields are characterized by the following theorem.

Theorem. A field T is a decomposition field if and only if it contains a subfield Z' isomorphic to Z. A field Λ is a splitting field if and only if it contains a subfield Γ isomorphic to the Galois field belonging to Z.

This follows directly from the definitions of splitting and decomposition fields in terms of absolutely irreducible representations. In particular, in analogy with the noncommutative case, one obtains the following corollary: a field Z' is a minimal decomposition field, i.e. no proper subfield is a decomposition field, if and only if it is isomorphic to Z. A minimal decomposition field is at the same time a splitting field if and only if Z is normal, hence a Galois field over P.

This is the reason for calling a simple algebra normal over its center. In a fixed algebraically closed Ω over P, there is only one minimal splitting field, namely the Galois field belonging to Z, in contrast to the generally infinitely many nonisomorphic fields in the noncommutative case. This has its source in the uniqueness of the direct decomposition in the commutative case, in contrast to the noncommutative case where uniqueness is only up to operator isomorphism; equivalently, in the finitely many different absolutely irreducible representations in the commutative case, as opposed to the infinitely many different ones in the noncommutative case, although they belong to the same class.

2. Hypercomplex proof of Galois theory in the case of separable fields Z/P. In exact analogy with §6, 3, the theory of isomorphisms holds for arbitrary Z separable over P; if Z is Galois, one can then pass to the usual automorphism group.

Hypotheses. The simple system Z over P is a finite separable extension field, Ω denotes a finite or infinite splitting field over P, $Z' = Z^{(1)}$ a subfield isomorphic to Z, and Γ the corresponding Galois field. By 1. there is a direct decomposition

$$Z_\Omega = r^{(1)} + \cdots + r^{(n)} = e^{(1)} Z_\Omega + \cdots + e^{(n)} Z_\Omega = e^{(1)}\Omega + \cdots + e^{(n)}\Omega.$$

The representation $Z \to Z^{(i)}$ generated by $e^{(i)}$, with $Z^{(i)} \subseteq \Gamma$, is defined by

$$e^{(i)} z = e^{(i)} z^{(i)}.$$

Lemma 1. The n representations generated by $e^{(1)}, \ldots, e^{(n)}$ are all distinct. The proof is as in §6, 3, or also follows from 1.; they belong to distinct representation classes.

Lemma 2. If T is an intermediate field between P and Z, and s is the rank of Z over T, then every isomorphism of T into Ω admits at least, and hence exactly, s extensions.

The proof is as in §6, 3. The strengthening of "at least" to "exactly" follows from the uniqueness of the decomposition of Z_Ω, according to which Z has not merely at least but exactly n isomorphisms in Ω.

As in §6, 3, one defines the partition $\mathfrak{I}_T$ induced by T of the finite set $\mathfrak{I}$ of isomorphisms of Z: precisely those isomorphisms in $\mathfrak{I}$ are regarded as equivalent in $\mathfrak{I}_T$ which induce the same isomorphism on T. One proves the following.

Main theorem. If $\mathfrak{I}_T$ is the partition induced by T, then T is a maximal subfield with respect to this partition.

From here, in the case of a Galois Z, one passes to the automorphism group and to the usual formulation exactly as in §6, 3 by composing with the reciprocal of an isomorphism. The corresponding assertion for subgroups follows from the consideration of idempotents, which is of interest in itself.

3. The idempotents of Z_Ω. Let S run through the n substitutions which carry Z into the individual $Z^{(i)}$, that is, the compositions of the isomorphisms $Z \to Z_1$ and $Z \to Z^{(i)}$, where the first is the reciprocal of $Z \to Z_1$. The field Z need not be Galois. For elements $a \in Z_\Omega$, the substitutions S are defined as operators by stipulating that they induce the identity on Z. Thus for each a the conjugate a^S lying in $Z_{(i)}$ is defined. With these conventions one has the following theorem.

Theorem. The n idempotents $e^{(i)}$ of Z_Ω are conjugate:

$$e^{(i)} = e^S, \qquad e = e^{(1)};$$

they lie in the n conjugate extension systems $Z\Omega$.

Indeed, applying S carries the defining relations

$$ez = ez^{(1)}, \qquad e^2 = e$$

to

$$e^S z = e^S z^{(i)}, \qquad (e^S)^2 = e^S.$$

By uniqueness of the decomposition the idempotents are themselves uniquely determined; hence $e^S = e^{(i)}$. Since Z is also a decomposition field, so that the representation $ez = ez^{(1)}$ is already mediated, e already lies in $Z\Omega$, and consequently e^S lies in $Z^S\Omega$.

If Z is Galois, the part of the main theorem of Galois theory concerning subgroups follows immediately. If $\mathfrak{H}$ is a subgroup of the Galois group $\mathfrak{G}$, then, by the uniqueness of the components of a direct sum, the element

$$\sum_{H \in \mathfrak{H}} e^H$$

admits all substitutions from $\mathfrak{H}$ and no others. Since the group $\mathfrak{G}$ was defined for Z, this is the Galois theory of Z; by isomorphism it also settles the theory for Z'.

4. Connection of the idempotents with complementary bases. Again Z is not assumed to be Galois.

Theorem. Let $a_1, \ldots, a_n$ be a basis of Z over P, and write the idempotent as

$$e = a_1\beta_1 + \cdots + a_n\beta_n,$$

so that

$$e^S = a_1\beta_1^S + \cdots + a_n\beta_n^S.$$

Then

$$\beta_1^S, \ldots, \beta_n^S$$

are the images, under the map mediated by e^S, of the complementary basis $b_1, \ldots, b_n$ to $a_1, \ldots, a_n$.

For the map mediated by e^S, $e^S x = e^S x^S$, in particular $e^S a_i = e^S a_i^S$, the relations

$$e^S e^S = e^S, \qquad e^S e^R = 0 \quad (S \neq R)$$

give the assignments $e^S \mapsto 1$, $e^R \mapsto 0$. Hence

$$1 = a_1^S\beta_1^S + \cdots + a_n^S\beta_n^S, \qquad 0 = a_1^R\beta_1^S + \cdots + a_n^R\beta_n^S \quad (S \neq R).$$

Written in matrix form, this is the defining equation for the complementary bases:

$$\begin{pmatrix} a_1^1 & \cdots & a_n^1 \\ \vdots & \ddots & \vdots \\ a_1^n & \cdots & a_n^n \end{pmatrix} \begin{pmatrix} \beta_1^S \\ \vdots \\ \beta_n^S \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}.$$

The passage to the trace definition follows because, for

$$a_i = \sum_S a_i^S e^S, \qquad b_i = \sum_S b_i^S e^S,$$

the matrix relation, after interchanging rows, becomes

$$\delta_{ij} = \mathrm{Sp}(a_i b_j).$$

Here the a_i and b_i lie in Z, since they admit all substitutions S.[5][6]

The contragredience of the complementary bases follows here from the fact that the idempotents e^S are invariants of Z, independent of the basis chosen. Thus all

$$a_1 \beta_1^S + \cdots + a_n \beta_n^S$$

are transformed into themselves when one passes to another basis

$$a_1', \ldots, a_n'$$

of Z/P.

§9. Splitting Fields and Decomposition Fields of Arbitrary Systems

1. Definitions and reduction to the simple case. The definitions given in §8 correspond in the general case to the following. Let S be hypercomplex over P. A commutative extension field Λ of P is called a splitting field of S if, in a composition series by one-sided ideals, S_Λ splits into absolutely simple composition factors; in other words, if all irreducible representations of S in Λ are already absolutely irreducible. An extension field T of P is called a decomposition field if S_T splits off at least one absolutely simple composition factor; equivalently, if at least one representation irreducible in T is already absolutely irreducible.

These definitions plainly contain the ones given in §8 for the commutative case and in §7 for simple algebras. The latter is true because A_Λ, for every commutative extension of the center, is completely reducible, so that composition factors and components of the direct sum decomposition coincide. Full matrix rings over the center, and only these, give absolutely simple components, hence also absolutely irreducible representations. Moreover A_Λ is two-sided simple, hence decomposes into operator-isomorphic components. Therefore the splitting off of one absolutely simple component already gives complete splitting: decomposition fields and splitting fields coincide.

From the definitions the following facts follow immediately: the splitting fields of S are the composita obtained by taking one splitting field for each representation irreducible in P; a decomposition field of S is any decomposition field of a representation irreducible in P. Because of the one-to-one correspondence between the simple systems, namely the components of the residue-class ring modulo the radical, and the representations irreducible in P, it is enough to consider simple systems. The results will then follow by combining §7 and §8.

2. Splitting fields and decomposition fields of simple systems. We first consider the case of a simple system A with separable center Z over P. From §8, 1 and §8, 3 one obtains the following. The two-sided decomposition of A_Ω corresponding to the direct decomposition of Z_Ω, where Ω denotes a splitting field of Z, gives n conjugate components

$$e^S A_\Omega$$

of A_Ω, each isomorphic to A, which become simple algebras over their centers

$$e^S Z_\Omega = e^S Z.$$

The substitutions S are defined as operators for A_Ω by stipulating that they induce the identity on A.

Indeed, by §8, 3 the idempotents are conjugate; the same is therefore true of the components $e^S A_\Omega$, by the definition of the substitutions for A. Since A is two-sided simple, $e^S A_\Omega$ is ring-isomorphic to A.

[5] Compare Representation Theory §25.

[6] Connections between complementary bases and idempotents occur first in Dedekind, Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen; compare vol. II of the collected works, pp. 4–5.

Moreover $e^S A_\Omega = e^S A_{Z^S}$, because $e^S Z_\Omega = e^S Z$; hence the center coincides with the coefficient domain, and the components are normal algebras.

The results of §7, 2 now further imply: if A is a simple system over P with separable center Z over P, then A_Ω too is without radical, hence completely reducible; here Ω may be any finite or infinite extension field. For by §7, 2, $e^S A_\Omega$ remains simple after every extension of coefficients, and therefore has no radical. Hence the same holds for the direct sum; this is A_Ω, or, if Ω is not a splitting field of Z, is obtained by coefficient extension from A_Ω.

From §7, 2 follows also the following characterization.

Characterization of decomposition fields and splitting fields. If A is a division algebra with separable center Z, then the irreducible embeddings of the decomposition fields are given by all and only the maximal commutative subfields of the A_f containing Z. The splitting fields are all and only the composita of a decomposition field with a splitting field of the center, in particular with the Galois field belonging to the center. A minimal decomposition field may be a splitting field even when the center is not Galois.

A decomposition field must contain a field isomorphic to Z; the assertion about embeddings of decomposition fields now follows from the preceding facts and from §7, 2. A splitting field must moreover contain a splitting field Ω of Z. But every decomposition field over Ω is at the same time a splitting field, since the components $e^S A_\Omega$ are simple and have a center isomorphic to Ω. If Z is Galois, minimal decomposition fields and splitting fields therefore coincide. But even in the non-Galois case, unlike the commutative case, there may be minimal decomposition fields which are at the same time splitting fields, namely whenever such a minimal decomposition field contains the Galois field belonging to Z. Example: a quaternion division algebra with center isomorphic to $P(\sqrt{2})$, P the field of rational numbers; the Galois field belonging to $P(\sqrt{2})$ is a minimal decomposition and splitting field.

Finally, if the center is inseparable, then Z_Ω, and with it A_Ω, is a system with radical. Let $\mathfrak{C}$ be the radical of Z_Ω, and let $\mathfrak{C}A_\Omega$ be the extension ideal in A_Ω. Then $Z_\Omega/\mathfrak{C}$ has a direct decomposition into conjugate components, in exact analogy with §8, 2. This gives, just as above, the direct decomposition of $A_\Omega/\mathfrak{C}A_\Omega$, in such a way that the components $e^S A_\Omega$ are isomorphic to A, with center $e^S Z^\Omega$. Thus the theorems on decomposition and splitting fields remain valid. The count of ranks further shows that the composition factors, corresponding to a composition series of $\mathfrak{C}A_\Omega$, are operator-isomorphic to $A_\Omega/\mathfrak{C}A_\Omega$, not merely homomorphic.

Expressed for irreducible representations, this gives the summary: if a representation of a hypercomplex system irreducible over P is given, then the representation is absolutely completely reducible if and only if the associated center is separable over P. In every case, the splitting and decomposition fields of the representation can be characterized as above by embedding into the associated simple system. In particular, the lowest degree of a decomposition field is the product of the degree of the center and the index of the associated algebra class over the center; the degree of every decomposition field is a multiple of this. The representation decomposes, in the sense of composition series, into as many distinct but conjugate absolutely irreducible representation classes as the degree of the largest separable field contained in the center. Each class occurs as many times as is given by the product of the index and the degree of the center over this separable extension.

For a perfect ground field, where only separable extension fields occur, this returns the known results of I. Schur, with the addition of the embedding characterization which already occurs in R. Brauer.

(Received 8 June 1932.)